

Midterm Exam: Micro 629 - Fall 2010

Texas A&M University

October 11th, 2010

By writing down your name you pledge to obey TAMU Honor Code:

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

Name (in pen, not pencil): _____

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
- Choose one problem from 1 and 2. Please state here which problem you are solving. (If you do not state it here, you will receive a score of zero from your work on these problems.) I am solving problem No: _____ (Use pen, not pencil here)
- No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.
- Some of the numerals in the different problems require to prove the preceding numerals in the problem; if you cannot prove one, move on, and assume that it is true; use it to prove what is next and then come back if you have time.

1. (20/100) Let $(\mathcal{B}, C)$ be a choice structure.

(a) (5/20) Formally state WARP.

(b) (15/20) Assume that $(\mathcal{B}, C)$ is a Walrasian choice structure, that is, $\mathcal{B} = \mathcal{B}^W$. Suppose that C is single-valued. Let x be the associated demand function. Prove that if $(\mathcal{B}^W, C)$ satisfies WARP then x satisfies the following property: for any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$.

2. (20/100) Let $(\mathcal{B}, C)$ be a choice structure.

(a) (5/20) Formally state WARP.

(b) (15/20) Assume that $(\mathcal{B}, C)$ is a Walrasian choice structure, that is, $\mathcal{B} = \mathcal{B}^W$. Suppose that C is single-valued. Let x be the associated demand function. Prove that if for any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$, then $(\mathcal{B}^W, C)$ satisfies WARP.

3. (20/100) Consider the preferences on $[1, 2] \subseteq \mathbb{R}$ that are represented by the functions in the following figure.

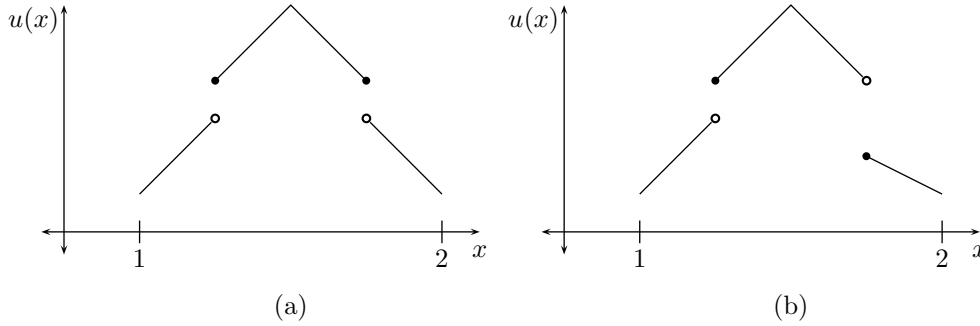


Figure 1: Utility functions in Problem 3

- (a) Is the preference in Figure 1 (a) continuous? (10/20) Formally support your answer.
 (b) Is the preference in Figure 1 (b) continuous? (10/20) Formally support your answer.
4. (40/100) Consider the preference on $\mathbb{R} \times \mathbb{R}_+$ represented by the following function: for each $(x_1, x_2) \in \mathbb{R} \times \mathbb{R}_+$

$$u(x_1, x_2) = x_1 + \varphi(x_2),$$

where φ is a continuous real-valued function.

Assume that for each $(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+$ there is a **unique** bundle that maximizes u in $B(p, w)$. Let x be the demand function associated with the preference represented by u . For the sake of simplicity you can normalize prices so the price of the first commodity is always 1.

- (a) (5/40) State the definition of quasi-linearity with respect to the first commodity (for preferences on $\mathbb{R} \times \mathbb{R}_+$)
 (b) (5/20) Prove that the preference represented by u is quasi-linear.
 (c) (7.5/40) Let $(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+$ and $w' \in \mathbb{R}_+$ such that $w' > w$. Prove that $x_2(p, w') = x_2(p, w)$. (Here a contradiction works fine, i.e., suppose that $x_2(p, w') \neq x_2(p, w)$,...)
 (d) (5/40) State the Compensated Law of Demand.
 (e) (5/40) Does x satisfy the Compensated Law of Demand? (here I am not asking for a proof; you should be able to give a precise answer, i.e., “yes because...” or “no, because....”)
 (f) (5/40) State the Uncompensated Law of Demand.
 (g) (7.5/40) Prove that x satisfies the Uncompensated Law of Demand.
5. (20/100) Let X be a finite set and B a binary relation on X . The **transitive closure** of B is the binary relation on X , denoted T_B , defined as follows: for each pair $\{x, y\} \subseteq X$, $x T_B y$ if and only if there is a list of elements of X , $\{x_i\}_{i=1}^n$ with $n \geq 1$ such that $x_1 = x$, $x_n = y$, $x_1 B x_2$, $x_2 B x_3, \dots$, $x_{n-2} B x_{n-1}$, and $x_{n-1} B x_n$.
- (a) (5/20) Prove that $B \subseteq T_B$.
 (b) (10/20) State the definition of transitivity of T_B . Prove that T_B is transitive.
 (c) (5/20) Let C be a transitive binary relation on X that contains B . Prove that $T_B \subseteq C$.